Project Report: SQL + Python Customer Churn Analysis - Data Analyst

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Family:	Data Analyst
Level:	Intermediate
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Query PostgreSQL warehouse, join customer data in pandas, analyze churn drivers, present to stakeholders.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Intermediate]

Chapter focus: Project Overview *(Level: Intermediate)**

SQL + Python Customer Churn Analysis is a intermediate-level end-to-end project for Data Analyst. Duration: 3 weeks. Query PostgreSQL warehouse, join customer data in pandas, analyze churn drivers, present to stakeholders. Tools: PostgreSQL, Python, Jupyter, PowerPoint.**

Code Reference:

```
Objective: Write 10+ SQL queries with JOINs and window functions
Objective: Identify churn predictors
Objective: Build cohort analysis
Objective: Present findings with storytelling framework
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is
A project overview defines scope, tools, and success criteria.

When to use it
At project kickoff.

Where to use it
Data Analyst portfolios and CodeQuest project submissions.

Real use example
A candidate lists this project on a Data Analyst resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: SQL [Intermediate]

Chapter focus: Phase: SQL *(Level: Intermediate)**

Extract cohorts from warehouse. Complete these tasks: CTE queries; Window functions. Deliverable for this phase: SQL script file.

Code Reference:

```
- [ ] CTE queries
- [ ] Window functions
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase SQL is one step in the full project lifecycle.

When to use it
During week 1 of the project timeline.

Where to use it
Data Analyst agile sprints and coursework milestones.

Real use example
Team demo shows the SQL script file to the teacher.

Chapter 3: Phase: Python [Intermediate]

Chapter focus: Phase: Python *(Level: Intermediate)**

pandas analysis on extract. Complete these tasks: Feature correlation; Cohort retention. Deliverable for this phase: Jupyter notebook.

Code Reference:

```
- [ ] Feature correlation
- [ ] Cohort retention
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase Python is one step in the full project lifecycle.

When to use it
During week 2 of the project timeline.

Where to use it
Data Analyst agile sprints and coursework milestones.

Real use example
Team demo shows the Jupyter notebook to the teacher.

Chapter 4: Phase: Story [Intermediate]

Chapter focus: Phase: Story *(Level: Intermediate)**

Apply Storytelling with Data. Complete these tasks: 3 key charts; So what? narrative. Deliverable for this phase: Slide deck.

Code Reference:

```
- [ ] 3 key charts
- [ ] So what? narrative
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase Story is one step in the full project lifecycle.

When to use it
During week 3 of the project timeline.

Where to use it
Data Analyst agile sprints and coursework milestones.

Real use example
Team demo shows the Slide deck to the teacher.

Chapter 5: Phase: Present [Intermediate]

Chapter focus: Phase: Present *(Level: Intermediate)**

Mock stakeholder review. Complete these tasks: Q&A prep; Appendix tables. Deliverable for this phase: Recorded presentation.

Code Reference:

```
- [ ] Q&A prep
- [ ] Appendix tables
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase Present is one step in the full project lifecycle.

When to use it
During week 4 of the project timeline.

Where to use it
Data Analyst agile sprints and coursework milestones.

Real use example
Team demo shows the Recorded presentation to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Intermediate]

Chapter focus: Final Deliverables and Review *(Level: Intermediate)**

Submit all final deliverables: SQL scripts; Jupyter notebook; Slide deck; Presentation recording. Skills demonstrated: SQL, pandas, Statistics, Storytelling. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: SQL scripts
Deliverable: Jupyter notebook
Deliverable: Slide deck
Deliverable: Presentation recording
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is
Deliverable review confirms end-to-end project completion.

When to use it
Before deadline and teacher review.

Where to use it
CodeQuest project gallery and interviews.

Real use example
A student submits SQL scripts and passes the rubric.